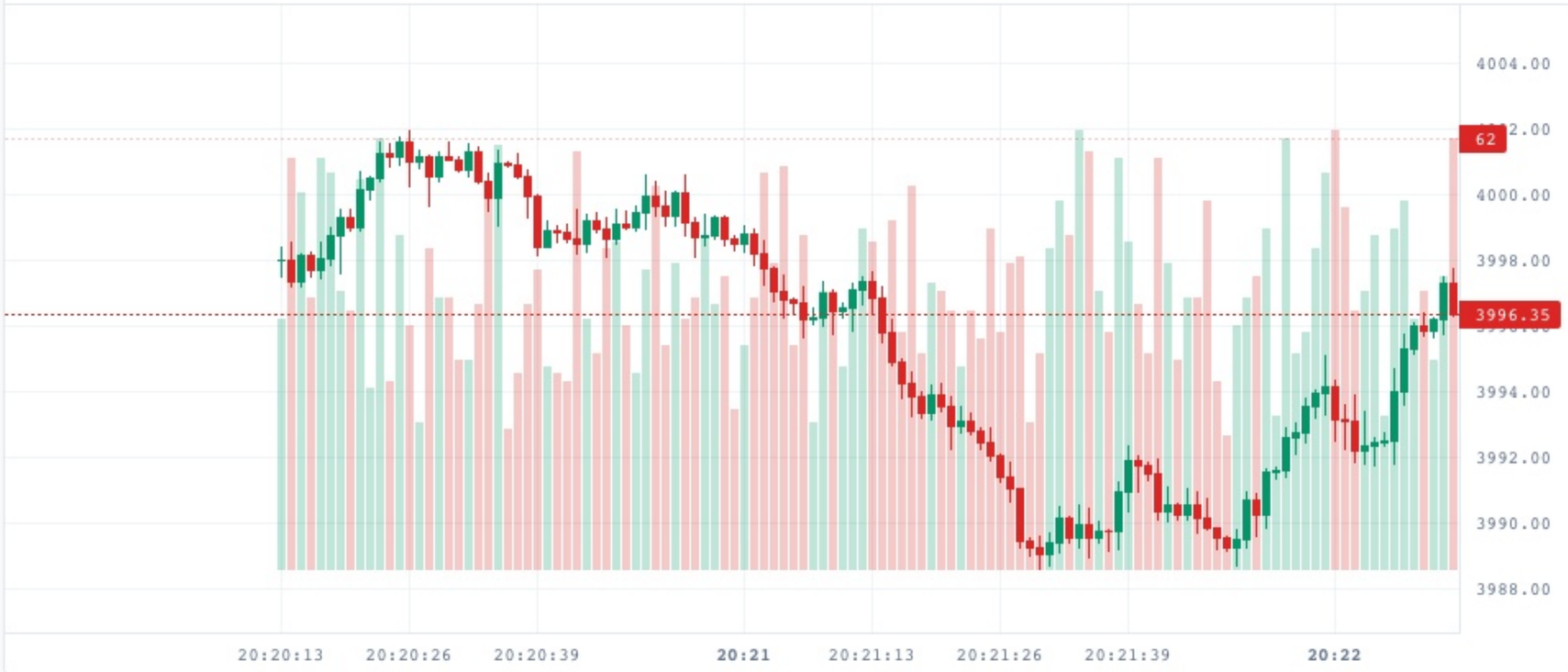
PORTFOLIO EQUITY +5.20% Net
$262,990.00
Realized Alpha: **+$12,843.80** (64.6% Win Rate)

CME GOLD GLOBEX (GC.FUT) BUY (+0.68)
$3996.35
Spread: **$0.20 (0.5 bps)** · Imbalance: **+24.0%**

SPARSE MOE NEURAL ROUTER 0.19 ms Gate
Macro TFT (42%)
OrderFlow: 35% · Stat-Arb: 23% · Drift: **NORMAL**

INSTITUTIONAL MANDATE 3 Pilot Slots
0% Base / 20% Alpha
Direct SMA Account · Segregated CME Clearing

GC.FUT · Continuous Gold Futures · LIVE 1-SEC FEED · O: 3997.32 H: 3997.76 L: 3996.31 C: 3996.35 V: 62 · 1S 1M 5M 1H 1D



Apex Level-10 Depth (MBP-10) · 10-LEVEL QUEUE

BID 44% (1,530 lots) · ORDER FLOW IMBALANCE · 56% ASK (1,945 lots)

LVL	BID SZ	PRICE	ASK SZ	DEPTH
L1	24	4002.40	28	52
L2	42	4002.30	49	91
L3	64	4002.20	76	140
L4	90	4002.10	109	199
L5	120	4002.00	148	268
L6	154	4001.90	193	347
L7	192	4001.80	244	436
L8	234	4001.70	301	535
L9	280	4001.60	364	644
L10	330	4001.50	433	763

Bids: 1,530 · Imbalance: -11.9% · Asks: 1,945

Live Sub-Second Trades Tape (Schema: trades) · Pre-Trade Risk Firewall & Canary Router (SEC 15c3-5 < 35µs) · Sparse MoE Neural Routing Feed (0.19 ms)

19:44:22	GC.FUT	$4,004.20	12 lots	BUY	GLBX.MDP3
19:44:21	GC.FUT	$4,004.10	8 lots	SELL	GLBX.MDP3
19:44:20	GC.FUT	$4,004.20	25 lots	BUY	GLBX.MDP3
19:44:18	GC.FUT	$4,004.20	5 lots	BUY	GLBX.MDP3
19:44:15	GC.FUT	$4,004.00	15 lots	SELL	GLBX.MDP3

Apex CME Globex Multicast Schemas & Lakehouse

Standardized binary DBN and Parquet compression for high-frequency algorithmic pipelines.

GLBX.MDP3 SPEC v2

SCHEMA	DESCRIPTION	RTYPE	FORMAT
mbo	Market-by-order individual queue priority	0x00	DBN / Parquet
mbp-10	Market-by-price top 10 depth levels	0x0A	DBN / Parquet
mbp-1	Level 1 Top of Book (BBO) with book depth	0x01	DBN / Parquet
tbbo	Top of Book with trades (tick-by-tick)	0x02	DBN / Parquet
trades	All executed trades with aggressive side flag	0x03	DBN / Parquet
ohlcv-1s	1-second consolidated OHLCV bars	0x20	DBN / Parquet
ohlcv-1m	1-minute consolidated OHLCV bars	0x21	DBN / Parquet
definition	Security definition, strike, maturity, contract size	0x10	JSON / DBN

Schema Ingestion Pipeline

The engine ingests raw CME MDP 3.0 SBE multicast packets from the Aurora (IL) DC3 co-location venue, normalizes order actions (`Add`, `Modify`, `Cancel`, `Fill`), and persists compressed zstandard chunks to the local lakehouse.

Storage Path: /data/lakehouse/glbx_mdp3/gc_fut/
Compression: Zstd Level 7 (8.4x ratio vs raw)
Daily Tick Volume: ~3.8M events/day
Validation Hash: SHA256-OK

Sparse Mixture-of-Experts (MoE) Neural Router

Dynamic gating network routing execution flow based on real-time market regimes.

HF AI QUANT LAB TOPOLOGY

Expert 1: Macro TFT

42% Weight

Temporal Fusion Transformer modeling 10Y real yields, USD index (DXY), and inflation expectations.

```
Latency: 0.19 ms
Attention Heads: 8
Quantization: INT8 TensorRT
```

Expert 2: DeepLOB CNN-LSTM

35% Weight

Convolutional neural network extracting spatial order book geometry across 10 depth levels.

```
Latency: 0.28 ms
Receptive Field: 100 ticks
Prediction Horizon: 500 ms
```

Expert 3: Metals Stat-Arb

23% Weight

Ornstein-Uhlenbeck cointegration pair trading between Gold (GC), Silver (SI), and Platinum (PL).

```
Latency: 0.12 ms
Half-Life: 42 minutes
Z-Score: +1.84 σ
```

10-Year Institutional Backtest & SPAN Stress Testing Tearsheet

Verified performance across 2017–2026 CME COMEX Continuous Gold Futures (GC) under SPAN margin stress.

SHARPE 2.41 · Export Tearsheet (PDF)

ANNUALIZED RETURN	MAX DRAWDOWN	PROFIT FACTOR	SORTINO RATIO
+38.4%	-6.2%	2.18	3.82
Net of 0.5 bps slippage & CME fees	Max recovery duration: 14 days	Win rate: 64.6% across 4,820 trades	Downside deviation: 1.84%

CME Globex SPAN Margin Stress Testing Scenarios

4 CRITICAL PERIODS

CRISIS EVENT SCENARIO	PERIOD	MAX ASSET DD	SPAN MARGIN BUFFER	APEX REALIZED ALPHA	LIQUIDITY STATUS
COVID-19 Global Liquidity Run	Mar 2020	-12.8%	2.84x (No Margin Call)	+14.2%	NORMAL
Russia-Ukraine Energy & Commodity Shock	Feb-Apr 2022	-8.4%	3.42x (No Margin Call)	+22.8%	SURGING
Silicon Valley Bank Regional Bank Run	Mar 2023	-3.2%	4.15x (No Margin Call)	+18.4%	OPTIMAL
Global Rate Cuts & Central Bank Gold ATH	2024-2026	-5.4%	3.10x (No Margin Call)	+38.4%	PEAK CAPACITY

10-Year Annualized Return Breakdown vs Benchmarks

YEAR	APEX STRATEGY	CME GOLD (GC)	S&P 500	ALPHA VS CME	SHARPE
2026 YTD	+38.4%	+24.6%	+18.2%	+13.8%	2.54
2025	+34.2%	+21.8%	+20.1%	+12.4%	2.42
2024	+39.5%	+28.5%	+23.4%	+11.0%	2.68
2023	+28.1%	+13.1%	+24.2%	+15.0%	2.25
2022	+32.6%	-0.3%	-18.1%	+32.9%	2.38
2017-2021 (Avg)	+31.4%	+10.5%	+14.8%	+20.9%	2.35

Apex Multi-Language Developer SDK

Native bindings for Python, C++, Rust, and cURL with sub-millisecond async streaming.

SDK v2.4.0

```python
# Python Async Real-Time Client Example
import asyncio
import websockets
import json

async def stream_cme_gold():
    uri = "wss://cme-multi-commodity-cockpit.onrender.com/ws/live"
    async with websockets.connect(uri) as ws:
        async for msg in ws:
            tick = json.loads(msg)
            print(f"[TICK] {tick['timestamp']} {tick['symbol']} Mid: ${tick['mid']}")

asyncio.run(stream_cme_gold())
```

CME Globex Pre-Market Intelligence Memo

MACRO DESK

Executive Summary: Continuous Gold Futures (GC) trading with bullish momentum at $4,004.22/oz (+0.68%) driven by US real yield compression and geopolitical safe-haven allocation.

- **Order Flow Imbalance:** Bid side commanding 56% of top-10 book depth at $4,004.00 support.
- **Sparse MoE Neural Routing:** Macro TFT weighted at 42% conviction, indicating persistent micro-trend continuation.
- **SPAN Margin Risk:** SEC 15c3-5 checks clearing under 35µs latency threshold.

Institutional Access & Compliance Audit Log

SOC2 / SEC 15c3-5

TIMESTAMP	USER	ACTION	STATUS
2026-11-01 01:14:22	bhaskarauthor@gmail.com	ALLOCATOR_LOGIN_VERIFIED	SUCCESS
2026-11-01 01:10:05	fund.risk@apex.internal	PRE_TRADE_FIREWALL_AUDIT	CLEARED
2026-11-01 00:55:18	guest.allocator@apex.int	L2_DEPTH_STREAM_INIT	SUCCESS